

Chapter 7: Approach Procedures & ILS Minima

Syllabus Source: CAE Oxford (pp. 173-206) | Pass Target: >= 90% AviationExam

1. THE FIVE APPROACH SEGMENTS

Under ICAO Doc 8168 (PANS-OPS), an instrument approach procedure is divided into up to five distinct segments. Each segment begins and ends at designated fixes, with specific obstacle clearance margins (MOC):

| Approach Segment           | Fix Boundaries                                | Operational Purpose & Minimum Obstacle Clearance (MOC)                                                                                                                                           |
|----------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Arrival Segment         | En-route airway to Initial Approach Fix (IAF) | Transitions aircraft from en-route ATS route to the terminal area. Often defined by a Standard Terminal Arrival (STAR). MOC conforms to en-route criteria.                                       |
| 2. Initial Segment         | IAF to Intermediate Fix (IF)                  | Provides initial descent and maneuvering to align with the intermediate approach. MOC is at least 300 m (984 ft) in the primary area. Reversal or racetrack procedures occur in this segment.    |
| 3. Intermediate Segment    | IF to Final Approach Fix (FAF / FAP)          | The segment where aircraft configuration and speed are adjusted for final approach. Optimum descent gradient is FLAT (max gradient 5.2% / 3.0°). MOC tapers from 300 m down to 150 m (492 ft).   |
| 4. Final Approach Segment  | FAF / FAP to Runway Threshold or MAPt         | Precision Approach: Guided by ILS/MLS glidepath to Decision Altitude/Height (DA/H). Non-Precision: Guided by VOR/NDB/LOC to Minimum Descent Altitude/Height (MDA/H).                             |
| 5. Missed Approach Segment | MAPt or DA/H to holding fix or en-route       | Executed if visual reference is not established. Divided into 3 PHASES: Initial (no turn permitted), Intermediate (climb at 2.5% gradient, MOC 30 m), and Final (climb to safe level, MOC 50 m). |

2. AIRCRAFT APPROACH CATEGORIES (A, B, C, D, E)

Aircraft categories are determined by Vat, which is the indicated airspeed at the runway threshold. Vat equals the stall speed in landing configuration (Vso) multiplied by 1.3, or 1.23 x Vs1g, at maximum certified landing mass:

| Category   | Vat Threshold Speed Range | Representative Aircraft Models                                      |
|------------|---------------------------|---------------------------------------------------------------------|
| Category A | Less than 91 kt           | Light singles and twins (Cessna 172, Piper PA-28, DA-42)            |
| Category B | 91 kt to 120 kt           | Heavy twins, regional turboprops (King Air, ATR-42, ATR-72, Dash-8) |
| Category C | 121 kt to 140 kt          | Medium commercial airliners (Airbus A320, Boeing 737, Embraer 190)  |
| Category D | 141 kt to 165 kt          | Heavy wide-body airliners (Boeing 777, 747, 787, Airbus A330, A350) |
| Category E | 166 kt to 210 kt          | Special high-performance military and experimental aircraft         |

### 3. REVERSAL & RACETRACK MANEUVERS

- 45°/180° Procedure Turn: Fly outbound on reversal track, turn 45° off track, fly for 1 MINUTE (Categories A and B) or 1 MIN 15 SEC (Categories C, D, and E), then execute a 180° turn in the opposite direction to intercept inbound track.
- 80°/260° Procedure Turn: Fly outbound on reversal track, turn 80° off track, immediately followed by a 260° turn in the opposite direction to intercept the inbound track.
- Base Turn: Aircraft flies outbound on a specified radial or dead reckoning track, then executes a turn to intercept the final approach track.
- Racetrack Procedure: Aircraft flies outbound parallel to inbound track for a specified time (1 to 3 minutes), then makes a turn to intercept the inbound track.

### 4. EXHAUSTIVE ILS PRECISION APPROACH MINIMA (CAT I TO CAT IIIC)

Precision approaches provide continuous electronic glidepath and azimuth guidance. Official EASA minima:

| ILS Category             | Decision Height (DH)                               | Minimum Runway Visual Range (RVR)                         |
|--------------------------|----------------------------------------------------|-----------------------------------------------------------|
| Category I (Cat I)       | Not lower than 200 ft (60 m)                       | Not less than 550 m RVR (or 800 m visibility without RVR) |
| Category II (Cat II)     | Lower than 200 ft but not lower than 100 ft (30 m) | Not less than 300 m RVR                                   |
| Category IIIA (Cat IIIA) | Lower than 100 ft or NO Decision Height            | Not less than 175 m RVR                                   |
| Category IIIB (Cat IIIB) | Lower than 50 ft or NO Decision Height             | Less than 175 m but not less than 50 m RVR                |
| Category IIIC (Cat IIIC) | NO Decision Height                                 | NO Runway Visual Range limitations (Zero-Zero)            |

#### [!] EXAM TRAP (AVIATIONEXAM) - DA/H vs MDA/H and the Missed Approach Initiation

- Decision Altitude/Height (DA/H): Used in PRECISION approaches. If visual reference is not established upon reaching DA/H, an IMMEDIATE MISSED APPROACH must be initiated. The aircraft may dip slightly below DA/H during the go-around.
- Minimum Descent Altitude/Height (MDA/H): Used in NON-PRECISION approaches. The aircraft MUST NEVER descend below MDA/H without visual reference. Level flight at MDA is permitted until reaching the Missed Approach Point (MAPt).

### 5. CONTINUOUS DESCENT FINAL APPROACH (CDFA)

CDFA is a technique for flying the final approach segment of a non-precision instrument approach procedure as a continuous descent, without level-offs, from an altitude/height at or above the FAF to a point approximately 15 m (50 ft) above the landing runway threshold or to the point where the missed approach is initiated. Under EASA Air Operations (Part-CAT), CDFA is mandatory for non-precision approaches unless approved otherwise.

### 6. AVIATIONEXAM REAL EXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

**Q1: What is the minimum Runway Visual Range (RVR) for an ILS Category I approach?**

[A] 300 m [B] 550 m [C] 800 m [D] 175 m

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. Under EASA CS-AWO and Part-CAT, the standard minimum RVR for an ILS Category I precision approach is 550 m (or 800 m flight visibility if no RVR system is available).

**Q2: In a precision approach, what must be done if visual reference is not established at the Decision Altitude/H**

[A] Level off and fly to the MAPt before executing go-around. [B] Initiate an immediate missed approach. [C] Descend an additional 50 ft to search for approach lights. [D] Revert to localizer-only non-precision minimums.

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. At DA/H on a precision approach, an immediate go-around must be initiated without delay if visual reference is not established. Level flight at DA/H is strictly prohibited.

**Q3: What Vat speed range corresponds to Aircraft Category C?**

[A] Less than 91 kt [B] 91 kt to 120 kt [C] 121 kt to 140 kt [D] 141 kt to 165 kt

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [C]. Category C Vat is 121 to 140 kt (representative of Airbus A320 and Boeing 737 at maximum landing mass).

**Q4: What is the Minimum Obstacle Clearance (MOC) provided in the intermediate approach segment?**

[A] 300 m (984 ft) [B] 150 m (492 ft) [C] 90 m (295 ft) [D] 50 m (164 ft)

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. In the intermediate segment, MOC is 150 m in the primary area, reducing from the 300 m MOC of the initial approach segment.

**Q5: During a 45°/180° procedure turn for a Category C aeroplane, how long must the outbound leg be flown aft**

[A] 1 minute [B] 1 minute 15 seconds [C] 1 minute 30 seconds [D] 2 minutes

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. PANS-OPS specifies 1 min for Categories A and B, and 1 min 15 sec for Categories C, D, and E.

**Q6: What is the standard design climb gradient for the intermediate and final phases of a missed approach?**

[A] 2.0% [B] 2.5% [C] 3.3% [D] 5.0%

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. Standard missed approach design climb gradient under PANS-OPS is 2.5%.

**Q7: Under Category II ILS operations, what is the allowable Decision Height range?**

[A] 200 ft to 150 ft [B] Lower than 200 ft but not lower than 100 ft [C] Lower than 100 ft but not lower than 50 ft [D] No Decision Height

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. Category II Decision Height is between 100 ft and 200 ft (RVR not less than 300 m).